

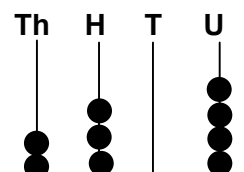
CPE 2006
MATHEMATICS

SECTION A (60 MARKS)

1. Write in words: 350

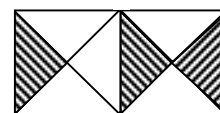
Answer :

2. What number is represented on the abacus ?



Answer :

3. What fraction of the figure is shaded ?



Answer :

4. Calculate: 327

$$+ 148$$

Answer :

5. Calculate: 894

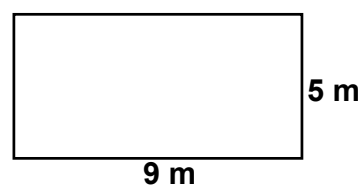
$$- 536$$

Answer :

6. Arrange 306, 630 and 603 in order starting with the largest.

Answer :

7. Find the area of the rectangle.

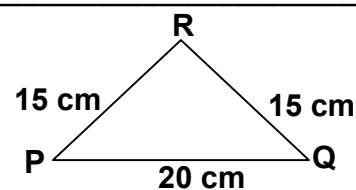


Answer : m^2

8. Calculate:
$$\begin{array}{r} 314 \\ \times 5 \\ \hline \end{array}$$

Answer :

9. Find the perimeter of triangle **PQR**.



Answer : cm

10. Calculate: $749 \div 7$

Answer :

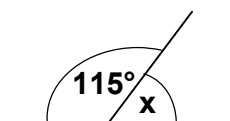
11. A cake costs Rs 7.50. Find the cost of 4 cakes.

Answer :

12. Convert 2 kg into grams.

Answer : g

13. Find angle x.



Answer :

14. Work out: $\frac{2}{3} + \frac{1}{6}$

Answer :

15. Calculate: 15% of 400

Answer :

For each question from numbers 16 to 45 draw a circle around the letter which shows the correct answer. An example has been done for you.

$$20 + 4 =$$

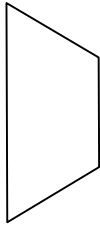
A 5

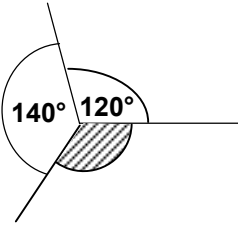
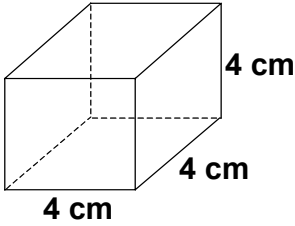
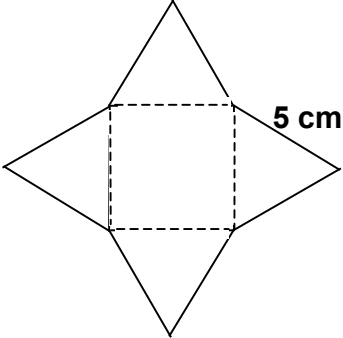
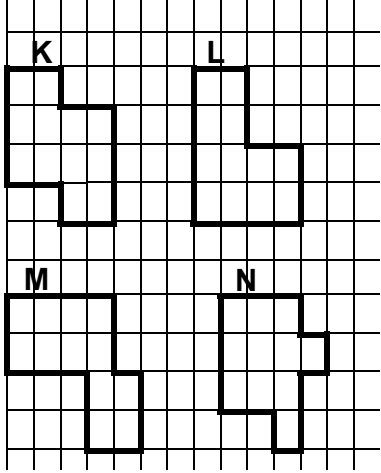
B 16

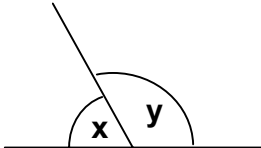
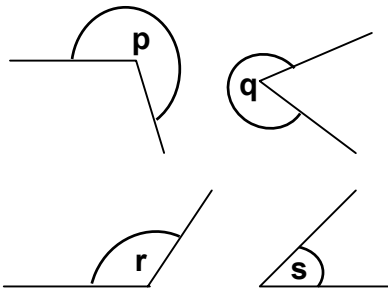
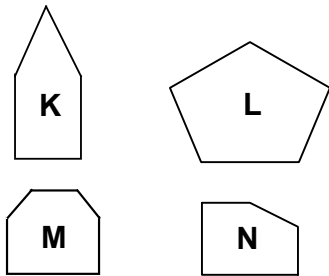
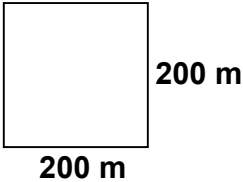
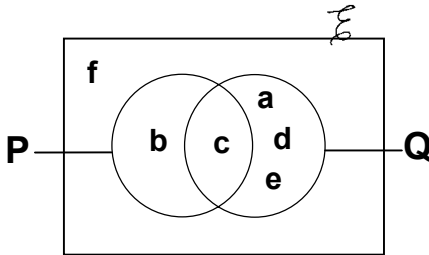
C 24

D 80

16. $(5 \times 100) + (2 \times 1000) + 6 + (7 \times 10) =$ A 2567 B 2576 C 5267 D 5276	17. What is the value of 3 in 3204 ? A 3 thousands B 3 hundreds C 3 tens D 3 units
18. Which one of the numbers 41, 51, 61, 71 is composite ? A 41 B 51 C 61 D 71	19. $3 \times 2^3 =$ A 6 B 8 C 12 D 24
20. Which one of the numbers 101, 201, 301, 401 is divisible by 3 ? A 101 B 201 C 301 D 401	21. $\frac{7}{10} \times \frac{3}{100} =$ A 0.021 B 0.21 C 2.1 D 21

22. Farida has 8 fifty-cent coins and 5 twenty-cent coins. The total value of these coins is ? A Rs 5 B Rs 13 C Rs 13.70 D Rs 500	23. Which one of the following fractions $\frac{4}{3}, \quad \frac{5}{4}, \quad \frac{6}{5}, \quad \frac{7}{6}$ is the largest ? A $\frac{4}{3}$ C $\frac{6}{5}$ B $\frac{5}{4}$ D $\frac{7}{6}$
24. 26 May 2004 was a Wednesday. 10 June 2004 was a A Tuesday. B Wednesday. C Thursday. D Friday.	25. What is the name of this shape ? A Kite B Parallelogram C Rectangle D Trapezium 
26. $6 \cdot 25 \times 10^3 =$ A 62.5 B 625 C 6250 D 62 500	27. There are 20 girls and 16 boys in a class. What fraction of the pupils are boys ? A $\frac{4}{9}$ C $\frac{5}{9}$ B $\frac{1}{2}$ D $\frac{4}{5}$
28. The set of all factors of 15 is A {3, 5} B {1, 15} C {1, 3, 5} D {1, 3, 5, 15}	29. Karan is 29 years 5 months old. Vima is 8 months younger than Karan. How old is Vima ? A 28 years 7 months B 28 years 9 months C 29 years 9 months D 30 years 1 month

30. Mr Tom sleeps for 6.5 hours every day. How many hours sleep does he have in a week ? A 30.5 hours B 32.5 hours C 42.5 hours D 45.5 hours	31. The height of Ashwin is 158 cm. Leena is 4 cm taller than Ashwin. Their average height is A 154 cm B 156 cm C 160 cm D 162 cm
32. The shaded angle is equal to A 100° B 120° C 140° D 260° 	33. The total surface area of the solid cube is A 16 cm^2 B 48 cm^2 C 64 cm^2 D 96 cm^2 
34. The figure is made up of a square and four equilateral triangles. Its perimeter is A 20 cm B 40 cm C 60 cm D 80 cm 	35. Which of the shapes K, L, M, N has the smallest area ? A K B L C M D N 
36. A rope is 3 m 50 cm long. It is cut into 2 equal parts. The length of each part is A 1 m 25 cm B 1 m 50 cm C 1 m 75 cm D 5 m 50 cm	37. A packet of biscuits weighs 275 g. What is the weight of 100 packets, in kilograms ? A 2.75 kg B 27.5 kg C 275 kg D 2750 kg

38. Ken has Rs 10. He buys a banana for Rs 1.50 and a cake for Rs 4.75. How much money has he left ? A Rs 3.75 B Rs 4.75 C Rs 5.75 D Rs 6.25	39. A box contains 50 beads. 16 of them are red, 13 are blue and the rest are green. What percentage of the beads are green ? A 21% B 29% C 42% D 58%
40. Angle y is twice angle x. Angle y = A 60° B 90° C 120° D 135° 	41. A tank, when half full, contains 8 L 65 cL of water. What is the capacity of the tank ? A 16 L 20 cL B 16 L 30 cL C 17 L 20 cL D 17 L 30 cL
42. Which one of the angles p, q, r, s is obtuse ? A p B q C r D s 	43. Which one of the shapes K, L, M, N is not a pentagon ? A K B L C M D N 
44. The area of the square is A 0.4 ha B 4 ha C 40 ha D 400 ha 	45. How many elements are not in set P ? A 4 B 3 C 2 D 1 

46. Write down the next term in each of the following sequences:

(i) 40, 70, 100, 130,

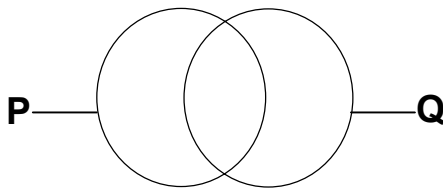
(ii) 91, 78, 65, 52,

(iii) (3 , 9), (5 , 17), (7 , 25), (9 , 33), (..... ,)

47. Brian's monthly salary is Rs 12 800. He gets an increase of 20%. What is his new salary ?

Answer :

48. $P = \{a, b, c, d\}$ and $Q = \{a, d, e\}$. Complete the Venn diagram below.



49. Jim bought three pens each costing the same price. He also bought two rulers each costing Rs 6.75. The total cost was Rs 42. Find the cost of a pen.

Answer :

50. Reena has three times as many stamps as Navin. Together they have 240 stamps. How many stamps does Reena have ?

Answer:

SECTION B (40 MARKS)

51. (a) The sum of two numbers is 8.05.

One of them is 3.57.

Calculate the other number.

Answer :

- (b) In a school there are 560 pupils.

$\frac{4}{7}$ of them are boys.

Of all the girls, 30% play badminton.

How many girls play badminton ?

Answer :

52. (a) Peter changes \$ 800 into rupees.
He spends $\frac{3}{4}$ of it.
How much money, in **rupees**, has he left ?

$$[\$ 1 = \text{Rs } 30.27]$$

Answer :

- (b) The **average** weight of two parcels is 5 kg 750 g. The **lighter** parcel weighs 4 kg 625 g. Calculate the weight of the **heavier** parcel.

Answer :

53. (a) The time in New York is G.M.T – 5 hours.
The time in Perth is G.M.T + 8 hours.

It is 18 00 on Monday in New York. What is the time and day in Perth ?

Answer : Time in Perth

Day in Perth

- (b) At a car park, the cost for parking is as follows:

For up to four hours there is a fixed cost of Rs 30.

For **each additional hour** after the first 4 hours the cost is Rs 20.

- (i) Carol parks her car for 2 hours. How much does she pay ?

Answer :

- (ii) Enrico parks his car for 6 hours. How much does he pay ?

Answer :

- (iii) Mike parks his car at 09 00. He pays a total of Rs 90 for parking.
What is the **latest** time by which he must take his car from the car park ?

Answer :

54. (a) A car travels 108 km in $1\frac{1}{2}$ hours.

A lorry travels 16 km in 20 minutes.

What is the ratio of the speed of the car to the speed of the lorry ?

Answer :

(b) A rectangular tank, which is 50 cm long, 30 cm wide and 30 cm high, contains 24 L of water.

When a stone is completely immersed in the water, the water level rises to 22 cm.

Calculate the volume of the stone in cm^3 .

Answer :

55. In 2003, a person working in a shop was paid Rs 250 per day from Monday to Saturday. To work on a Sunday he was paid $1\frac{1}{2}$ times as much. He worked every day from Wednesday 10 December 2003 to Tuesday 23 December 2003.

(i) How much money did he earn in total ?

In 2004, the amount of money paid to the same person to work on a Sunday increased. The amount paid for other days was unchanged. He earned Rs 5500 for working every day from Friday 20 February 2004 to Tuesday 9 March 2004.

(ii) Calculate the percentage increase in the amount paid to the worker to work on a Sunday.

Answer : (i)

Answer : (ii)

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